

TOTALLY INVARIANT SUBVARIETIES OF PROJECTIVE SPACE

GPT 5.6 PRO, COMMUNICATED BY YUJIE LUO AND SHENG MENG

ABSTRACT. Let $f : \mathbf{P}_{\mathbb{C}}^n \rightarrow \mathbf{P}_{\mathbb{C}}^n$ be a surjective endomorphism of algebraic degree greater than one. We prove that every integral closed subvariety V with $f^{-1}(V) = V$ set-theoretically is a linear subspace.

Disclosure. We claim no credit for this work. We are sharing it solely to present a proof generated by GPT 5.6 Pro. We have simplified and verified the proof. More intuitive versions are being developed and are available [online](#).

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1. INTRODUCTION

We work over $\mathbb{C}$. Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be a surjective endomorphism of algebraic degree $q > 1$, so that $f^*\mathcal{O}_{\mathbf{P}^n}(1) \simeq \mathcal{O}_{\mathbf{P}^n}(q)$. A closed subvariety $V \subseteq \mathbf{P}^n$ is *totally invariant* under f if $f^{-1}(V) = V$ as sets. Such subvarieties arise as exceptional loci in backward dynamics. Coordinate linear subspaces are examples for the power map

$$[x_0 : \cdots : x_n] \mapsto [x_0^q : \cdots : x_n^q].$$

The following classical conjecture predicts that every integral totally invariant subvariety is linear; see [19, arXiv version, Conjecture 1.4].

Linearity Conjecture. *Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be a surjective endomorphism of algebraic degree greater than one, and $V \subseteq \mathbf{P}^n$ an integral closed subvariety. If $f^{-1}(V) = V$ set-theoretically, then V is a linear subspace.*

For curves in $\mathbf{P}^2$, Fornæss–Sibony established degree bounds that left only a smooth conic as a possible nonlinear example [12, Proposition 4.2 and Section 4, p. 212]. Cerveau–Lins Neto ruled out this possibility and proved the plane-curve case [8, Theorem 2 and Sections 3–4]. Gurjar later gave another

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proof as a consequence of a ramification theorem [14]; see also [6, Remark 5]. Cerveau–Lins Neto and Beauville also settled the conjecture for smooth hypersurfaces in arbitrary dimension [8, 1]. The remaining cases therefore concern singular hypersurfaces and subvarieties of higher codimension.

Briend–Cantat–Shishikura announced a proof in arbitrary dimension in 2004 [2]. Their argument relies on a local-degree lemma of Briend–Duval [3, Section 2], whose proof Dinh found to be incorrect in 2006. The erratum [4] repairs Briend–Duval’s equidistribution argument, but the gap in the linearity proof remains; Cantat–Xie explain this distinction in [6, Remark 6].

For singular hypersurfaces, D.-Q. Zhang reduced the nonlinear case in $\mathbf{P}^3$ to four possible singular cubic surfaces [22, Theorem 1.1]. Horing then proved that every totally invariant prime divisor in $\mathbf{P}^3$ is a hyperplane [17, Corollary 1.2 and its proof], using a logarithmic-differential estimate together with earlier results, including Nakayama–Zhang’s exclusion of degree two [19, arXiv version, Section 5]. Mabed extended this method to irreducible totally invariant hypersurfaces with at most isolated singularities in $\mathbf{P}^n$, for every $n \geq 2$ [18, Corollary 1.4]. More recently, Broustet–Horing proved linearity for all totally invariant prime divisors in $\mathbf{P}^4$ and for normal totally invariant hypersurfaces in every dimension [5, Corollaries 1.5 and 1.3].

In higher codimension, Chen–Meng–Zhong proved that a smooth rationally connected projective threefold of Picard number two admitting an int-amplified endomorphism is toric [10, Theorem 1.3]. Their result implies that every blowup-equivariant smooth curve in $\mathbf{P}^3$ for a non-isomorphic surjective endomorphism is a line [10, Corollary 1.6]. In arbitrary codimension, Chang–Luo established sharp bounds for the total degrees of subvarieties that are totally invariant under iterates of a fixed endomorphism [9, Theorem 1.3].

We prove the following theorem, which establishes the linearity conjecture in full.

Theorem 1.1. *Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be a surjective endomorphism of algebraic degree greater than one, and let $V \subseteq \mathbf{P}^n$ be an integral closed subvariety. If $f^{-1}(V) = V$ set-theoretically, then V is a linear subspace.*

Theorem 1.1 imposes no smoothness or normality assumption on V . It also holds over any algebraically closed field of characteristic zero: one can descend f and V to a field finitely generated over $\mathbb{Q}$ and embed that field into $\mathbb{C}$ [21, Tag 01ZM]. Both total invariance and linearity can be checked after extending the ground field. The distinction between set-theoretic and scheme-theoretic inverse images is central to the proof: a general fiber of the restriction to V is reduced, while the corresponding fiber in $\mathbf{P}^n$ need not be.

Here is the argument. Write $r = \dim V$ and $\delta = \deg V$, and assume $1 \leq r < n$. The trace splitting of [9], together with iteration, gives

$$H^i(\mathbf{P}^n, \mathcal{I}_V(t)) = 0 \quad (i > 0, t > 0).$$

Thus, for every $t > 0$, sections of $\mathcal{O}_V(t)$ extend to $\mathbf{P}^n$, and $H^i(V, \mathcal{O}_V(t)) = 0$ for all $i > 0$.

We may replace f by an iterate and continue to write f and q for the map and its algebraic degree. Put $g = f|_V$. A general fiber $S = g^{-1}(y)$ consists of q^r distinct smooth points. A relative Jacobian calculation produces a single polynomial, of degree at most $(n - r)(q - 1)$, whose class generates the socle of the ambient fiber algebra at each point of S . Multiplying by this polynomial turns local residues into nonzero multiples of point evaluation. Euler–Jacobi vanishing then gives a relation

$$\sum_{x \in S} \lambda_x \sigma(x) = 0, \quad \lambda_x \neq 0, \quad \sigma \in H^0(V, \mathcal{O}_V(rq - r - 1)).$$

Values are taken in a fixed trivialization along S .

If $\delta > 1$, a general complementary linear section of V contains at least two points. Its inverse image under g is a reduced complete intersection on V . Extend the relation on one fiber by zero on the remaining fibers. Complete-intersection duality lifts the resulting relation to a nonzero section of $\omega_V(r + 1)$ vanishing on another entire fiber. Since the canonical sheaf ω_V is torsion-free of rank one, there is an injection

$$\omega_V(r + 1) \hookrightarrow \mathcal{O}_V(m)$$

for some $m > 0$ depending only on $V \subseteq \mathbf{P}^n$. The resulting section of $\mathcal{O}_V(m)$ vanishes at all q^r points of a fiber $g^{-1}(y')$. Intersecting its zero divisor with the pullbacks of $r - 1$ general target hyperplanes through y' gives

$$q^r \leq mq^{r-1}\delta.$$

Choosing the iterate so that $q > m\delta$ gives a contradiction, so $\delta = 1$.

Section 2 establishes the cohomology vanishing. Sections 3 and 4 construct the fiber relation, Section 5 proves the lifting statement, and Section 6 completes the degree comparison.

For a closed subscheme $X \subseteq \mathbf{P}^n$, we write $\mathcal{O}_X(t) = \mathcal{O}_{\mathbf{P}^n}(t)|_X$ and, for a coherent sheaf $\mathcal{F}$ on X , $\mathcal{F}(t) = \mathcal{F} \otimes \mathcal{O}_X(t)$. The smooth locus of V is denoted by V_{reg} , and the dual of a finite-dimensional vector space E by $E^\vee$.

2. TRACE AND COHOMOLOGY VANISHING

We first record the degrees of the ambient map and its restriction.

Lemma 2.1. *Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be a surjective endomorphism of algebraic degree q . Then f is finite flat of degree q^n . If V is an integral totally invariant subvariety of dimension r , then $g = f|_V$ is finite surjective and*

$$g^* \mathcal{O}_V(1) \simeq \mathcal{O}_V(q), \quad \deg g = q^r.$$

There is a dense open set $U \subseteq V_{\text{reg}}$ such that $g^{-1}(U) \subseteq V_{\text{reg}}$ and $g^{-1}(U) \rightarrow U$ is finite étale.

Proof. A positive-dimensional fiber of f would contain a curve on which $f^*\mathcal{O}_{\mathbf{P}^n}(1) = \mathcal{O}_{\mathbf{P}^n}(q)$ has degree zero, contradicting ampleness. Thus f is proper and quasi-finite, hence finite. It is flat by miracle flatness, since source and target are smooth of the same dimension [21, Tag 00R4]. The projection formula gives $\deg f = q^n$.

Total invariance and surjectivity give $f(V) = V$. Since V is reduced, $f|_V$ factors through a finite surjective morphism $g : V \rightarrow V$. The polarization restricts to V , and the projection formula gives

$$(\deg g) \deg V = q^r \deg V.$$

Hence $\deg g = q^r$. In characteristic zero the restriction is generically étale. Removing its branch locus, $V \setminus V_{\text{reg}}$, and the image of $V \setminus V_{\text{reg}}$ gives U . $\square$

The next proposition is the trace argument of [9, Lemma 2.1] in projective space.

Proposition 2.2. *Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be a surjective endomorphism of algebraic degree $q > 1$, and let V be a reduced closed subscheme with $f^{-1}(V) = V$ set-theoretically. Then $\mathcal{I}_V(t)$ is a direct summand of $f_*\mathcal{I}_V(qt)$ for every integer t . Consequently,*

$$(2.1) \quad H^i(\mathbf{P}^n, \mathcal{I}_V(t)) = 0 \quad (i > 0, t > 0).$$

Proof. The finite flat morphism f has rank q^n , and its algebra trace commutes with base change [21, Tag 0BT8]. We claim that

$$\text{Tr}_f(f_*\mathcal{I}_V) \subseteq \mathcal{I}_V.$$

Indeed, let b be a local section of $f_*\mathcal{I}_V$ near a closed point $y \in V$. Every point of $f^{-1}(y)$ belongs to V . The image of b in the finite fiber algebra therefore vanishes at every point and is nilpotent. Multiplication by this image has trace zero, so $\text{Tr}_f(b)(y) = 0$. Since V is reduced, this proves the claim.

Pullback also preserves $\mathcal{I}_V$. The composite

$$\mathcal{I}_V \longrightarrow f_*\mathcal{I}_V \xrightarrow{\text{Tr}_f} \mathcal{I}_V$$

is multiplication by q^n . Dividing the trace by q^n gives a splitting. Tensoring with $\mathcal{O}_{\mathbf{P}^n}(t)$ and applying the projection formula proves the direct-summand assertion. Iterating the splitting shows that, for fixed $i, t > 0$, the group $H^i(\mathbf{P}^n, \mathcal{I}_V(t))$ is a direct summand of $H^i(\mathbf{P}^n, \mathcal{I}_V(q^k t))$ for every $k \geq 1$. Serre vanishing [16, Theorem III.5.2] makes the latter group zero for $k \gg 0$, proving (2.1). $\square$

Corollary 2.3. *Under the hypotheses of Proposition 2.2, for every $t > 0$ the restriction map*

$$H^0(\mathbf{P}^n, \mathcal{O}_{\mathbf{P}^n}(t)) \longrightarrow H^0(V, \mathcal{O}_V(t))$$

is surjective, and $H^i(V, \mathcal{O}_V(t)) = 0$ for all $i > 0$.

Proof. Apply (2.1) and the cohomology vanishing for $\mathcal{O}_{\mathbf{P}^n}(t)$ to

$$0 \longrightarrow \mathcal{I}_V(t) \longrightarrow \mathcal{O}_{\mathbf{P}^n}(t) \longrightarrow \mathcal{O}_V(t) \longrightarrow 0. \quad \square$$

3. A RELATIVE JACOBIAN LEMMA

For an Artinian local ring $(Q, \mathfrak{m}_Q)$, its socle is $\text{Soc}(Q) = \text{Ann}_Q(\mathfrak{m}_Q)$. The Jacobian theorem of Scheja–Storch states that, over a field of characteristic zero, the socle of a zero-dimensional complete intersection

$$Q = K[[z_1, \dots, z_c]]/(H_1, \dots, H_c)$$

is generated by the nonzero class of $\det(\partial H_a / \partial z_b)$; see [20] and [11, Theorems 7.1 and A.5]. We need a relative version in which parameters on the reduced space may be lifted arbitrarily to the thickening.

Lemma 3.1. *Let $B = \mathbb{C}[[s_1, \dots, s_r]]$, $R = B[[z_1, \dots, z_c]]$, and*

$$A = R/(H_1, \dots, H_c), \quad \Delta = \det \left(\frac{\partial H_a}{\partial z_b} \right) \in A,$$

where $r, c \geq 1$ and the H_a form a regular sequence. Assume that A is finite flat over B and that $A_{\text{red}} \simeq B$ as a B -algebra. If N is the nilradical of A , then

$$\text{Ann}_A(N) = B\Delta.$$

Moreover, if $\tau_1, \dots, \tau_r \in A$ reduce to a regular system of parameters of A/N , then the image of Δ generates the socle of $A/(\tau_1, \dots, \tau_r)$ and is nonzero.

Proof. Translate each z_b by its image in $A/N = B$. Then the kernel of $R \rightarrow A/N$ is $(z_1, \dots, z_c)$, and differentiation in the z_b with the s_i fixed is unchanged. Let K be the fraction field of B and put $\mathfrak{p} = (z_1, \dots, z_c) \subset R$. Since $\sqrt{(H_1, \dots, H_c)} = \mathfrak{p}$, the finite local K -algebra $A_K = A \otimes_B K$ identifies with $R_{\mathfrak{p}}/(H_1, \dots, H_c)$. Completing $R_{\mathfrak{p}}$ gives $K[[z_1, \dots, z_c]]$ and leaves its Artinian quotient A_K unchanged. This presents A_K as a zero-dimensional complete intersection with residue field K . The Jacobian theorem therefore gives $N_K \Delta = 0$. Since A is B -flat, $A \rightarrow A_K$ is injective, and hence $N\Delta = 0$ in A .

Put $M = \text{Ann}_A(N)$. The closed fiber $A/\mathfrak{m}_B A$ is an Artinian complete intersection and hence Gorenstein. Lift a linear functional defining its perfect multiplication pairing to a B -linear map $\ell : A \rightarrow B$. The A -linear map

$$A \longrightarrow \text{Hom}_B(A, B), \quad a \longmapsto (b \longmapsto \ell(ab)),$$

is an isomorphism modulo $\mathfrak{m}_B$ and hence an isomorphism between free B -modules of the same rank. Finite adjunction consequently gives an isomorphism of B -modules

$$M = \text{Hom}_A(B, A) \simeq \text{Hom}_A(B, \text{Hom}_B(A, B)) \simeq B.$$

The Jacobian theorem on the closed fiber gives $\Delta \notin \mathfrak{m}_B A$, and therefore $\Delta \notin \mathfrak{m}_B M$. As M is free of rank one over B , it follows that $M = B\Delta$.

Now let $B' = \mathbb{C}[[T_1, \dots, T_r]]$ act on A by $T_i \mapsto \tau_i$. The induced map $B' \rightarrow A/N$ is an isomorphism. Each quotient in the finite filtration by powers of N is finite over A/N , so A is finite over B' . The τ_i form a system of parameters in the Cohen–Macaulay ring A ; thus A is free over B' . The closed fiber $A/(\tau_1, \dots, \tau_r)$ is again a zero-dimensional complete intersection, so $\mathrm{Hom}_{B'}(A, B')$ is free of rank one over A by the same argument.

The quotient $A \rightarrow A/N = B'$ splits as a map of B' -modules. Dualizing gives a split injection $B' \hookrightarrow \mathrm{Hom}_{B'}(A, B')$ whose image consists of the functionals vanishing on N . Under an A -linear identification $\mathrm{Hom}_{B'}(A, B') \simeq A$, this image is precisely $\mathrm{Ann}_A(N) = M$. Thus $M \hookrightarrow A$ splits over B' . The actions of B and B' on M both factor through A/N , so $M = B'\Delta$. Reduction modulo $(T_1, \dots, T_r)$ preserves the split injection. Hence Δ remains nonzero in $A/(\tau_1, \dots, \tau_r)$. The maximal ideal of this quotient is the image of N and annihilates Δ . Its socle is one-dimensional, proving the assertion. $\square$

4. RELATIONS ON A FIBER

Let $p_1, \dots, p_n$ be polynomials in affine coordinates $\xi_1, \dots, \xi_n$, with an isolated common zero x . Write the local Grothendieck residue as

$$\rho_x(P) = \mathrm{Res}_x \left[\frac{P d\xi_1 \wedge \dots \wedge d\xi_n}{p_1 \cdots p_n} \right].$$

The pairing $(a, b) \mapsto \rho_x(ab)$ is perfect on the local complete-intersection algebra [7, Theorem 1.5.1]. If the highest homogeneous parts of the p_i have no common nonzero zero, Euler–Jacobi vanishing gives

$$(4.1) \quad \sum_x \rho_x(P) = 0 \quad \text{for} \quad \deg P < \sum_{i=1}^n (\deg p_i - 1).$$

The sum includes all common zeros, and the fiber algebras need not be reduced [7, Theorem 1.5.12].

Proposition 4.1. *Let $f : \mathbf{P}^n \rightarrow \mathbf{P}^n$ be a surjective endomorphism of algebraic degree $q > 1$, and let V be an integral totally invariant subvariety of dimension $1 \leq r < n$. Put $g = f|_V$ and $t = r(q - 1) - 1$. Suppose $y \in V_{\mathrm{reg}}$, g is étale over y , and $S = g^{-1}(y) \subseteq V_{\mathrm{reg}}$. After choosing a trivialization of $\mathcal{O}_V(t)$ along S , there are nonzero scalars λ_x , $x \in S$, such that*

$$(4.2) \quad \sum_{x \in S} \lambda_x \sigma(x) = 0 \quad \text{for every } \sigma \in H^0(V, \mathcal{O}_V(t)).$$

Proof. Put $c = n - r$. Choose target coordinates $[Y_0 : \dots : Y_n]$ with $y = [1 : 0 : \dots : 0]$, and write $u_i = Y_i/Y_0$. We may assume that $u_1, \dots, u_r$ restrict to parameters on V at y , while $du_{r+1}, \dots, du_n$ span its conormal space there. The completed ideal of V at y then has generators

$$h_a = u_{r+a} + O((u_1, \dots, u_n)^2), \quad 1 \leq a \leq c.$$

In particular, $u_1, \dots, u_r, h_1, \dots, h_c$ are ambient local parameters.

Write $f = [\Phi_0 : \cdots : \Phi_n]$, where the Φ_i are degree- q forms without a common zero. Choose a source hyperplane avoiding $f^{-1}(y)$ and not contained in any $\{\Phi_i = 0\}$ for $1 \leq i \leq n$. On its affine complement, let p_i be the dehomogenization of Φ_i , for $0 \leq i \leq n$. Then $\deg p_i = q$ for $1 \leq i \leq n$, and $p_1 = \cdots = p_n = 0$ defines the whole ambient fiber. Their highest homogeneous parts have no common nonzero zero: such a zero would give a point at infinity with $\Phi_1 = \cdots = \Phi_n = 0$ and $\Phi_0 \neq 0$, contrary to the choice of the hyperplane.

Choose affine linear coordinates $(s_1, \dots, s_r, z_1, \dots, z_c)$ such that the ds_i restrict to a basis of the cotangent space of V at every point of S . This is a nonempty open condition for each of finitely many points, so the choices can be made simultaneously. On this affine chart, define the polynomial

$$\Theta = \det \left(\frac{\partial p_{r+a}}{\partial z_b} \right)_{1 \leq a, b \leq c}.$$

It has degree at most $c(q-1)$. We claim that its class generates the socle of the local ambient fiber algebra at every $x \in S$.

Fix $x \in S$ and center the source coordinates at x . In the completed local ring, V is a graph $z_b = \varphi_b(s)$. Set $w_b = z_b - \varphi_b(s)$, so that differentiation in w_b agrees with differentiation in z_b when s is fixed. The function p_0 is a unit at x . Put

$$v_i = p_i/p_0 = u_i \circ f, \quad H_a = h_a(v_1, \dots, v_n), \quad A = \widehat{\mathcal{O}}_{\mathbf{P}^n, x} / (H_1, \dots, H_c).$$

To apply Lemma 3.1, let I and J be the ideals of V and $f^{-1}(V)$ in the uncompleted local ring at x . Total invariance gives $\sqrt{J} = I$, hence $I^e \subseteq J \subseteq I$ for some $e \geq 1$. After completion this becomes

$$(w_1, \dots, w_c)^e \subseteq (H_1, \dots, H_c) \subseteq (w_1, \dots, w_c).$$

Thus $A_{\text{red}} = B = \mathbb{C}[[s_1, \dots, s_r]]$. The monomials in the w_b of total degree less than e generate A over B , so A is finite over B . The c elements H_a generate an ideal of height c in a regular local ring and form a regular sequence. Consequently A is Cohen–Macaulay, and the parameters $s_1, \dots, s_r$ make it free over B .

Since g is étale at x , the reductions of $v_1, \dots, v_r$ are parameters of A_{red} . Apply Lemma 3.1 with normal variables w_b and parameter lifts $\tau_i = v_i$. It follows that the relative Jacobian

$$\Delta = \det \left(\frac{\partial H_a}{\partial w_b} \right) = \det \left(\frac{\partial H_a}{\partial z_b} \right)$$

has nonzero socle class in $A/(v_1, \dots, v_r)$. This quotient is the local algebra of $f^{-1}(y)$ at x , because $u_1, \dots, u_r, h_1, \dots, h_c$ are local parameters at y . Equivalently, it is the quotient of the completed source ring by $(p_1, \dots, p_n)$. In this quotient the chain rule gives

$$\frac{\partial H_a}{\partial z_b} \equiv \frac{\partial v_{r+a}}{\partial z_b} \equiv p_0^{-1} \frac{\partial p_{r+a}}{\partial z_b}.$$

Indeed, $H_a - v_{r+a} \in (v_1, \dots, v_n)^2$, so its derivatives lie in $(v_1, \dots, v_n)$. Differentiating $v_{r+a} = p_{r+a}/p_0$ gives the second congruence, since the additional term is divisible by p_{r+a} . Taking determinants yields

$$\Delta \equiv p_0^{-c} \Theta \pmod{(p_1, \dots, p_n)}.$$

Since p_0 is a unit, Θ also has nonzero socle class. The local fiber algebra is Artinian and equals its completion, so this formal calculation applies to the algebra on which ρ_x is defined. This proves the claim.

Using the fixed affine coordinates (s, z) , set $\lambda_x = \rho_x(\Theta)$. The socle property gives

$$\rho_x(\Theta P) = \lambda_x P(x) \quad \text{for every local function } P.$$

Moreover, $\lambda_x \neq 0$: otherwise the nonzero class of Θ would pair to zero with every element, contradicting the perfect residue pairing.

Since $c = n - r$ and $t = r(q - 1) - 1$, a polynomial P of degree at most t satisfies

$$\deg(\Theta P) \leq c(q - 1) + t = n(q - 1) - 1.$$

The support of the ambient fiber is exactly S , so (4.1) gives

$$0 = \sum_{x \in S} \rho_x(\Theta P) = \sum_{x \in S} \lambda_x P(x).$$

The chosen affine chart trivializes $\mathcal{O}_V(t)$ along S . For $t > 0$, every section of $\mathcal{O}_V(t)$ is represented in this chart by a polynomial of degree at most t , by Corollary 2.3. The only remaining case is $t = 0$, when every global section is constant because V is integral and projective. Thus (4.2) holds in all cases. $\square$

5. LIFTING RELATIONS BY DUALITY

Let $V \subseteq \mathbf{P}^n$ be integral of dimension r . Its canonical sheaf ω_V is the degree $-r$ cohomology sheaf of its normalized dualizing complex $\omega_V^\bullet$. It is a coherent torsion-free sheaf of rank one and agrees with the canonical line bundle on V_{reg} [21, Tags 0FVV and 0AXY]. In this section we use proper duality as in [15] and [21, Tag 0FVV].

The following form of complete-intersection duality turns a relation among point values into a global section. The common zero scheme lies in the smooth locus, but V itself need not be Cohen–Macaulay.

Proposition 5.1. *Assume $r \geq 1$ and $H^i(V, \mathcal{O}_V(j)) = 0$ for all $i, j > 0$. Let $q > r + 1$ be an integer, and let $\sigma_1, \dots, \sigma_r \in H^0(V, \mathcal{O}_V(q))$ have a reduced zero-dimensional common zero scheme $T \subseteq V_{\text{reg}}$. Put $t = rq - r - 1$.*

Let λ be a functional on $H^0(T, \mathcal{O}_T(t))$ annihilating the restrictions of all sections of $\mathcal{O}_V(t)$. Then λ is induced by a section

$$a \in H^0(V, \omega_V(r + 1)),$$

through duality on T and adjunction for the ordered sections σ_i . Under the decomposition

$$H^0(T, \mathcal{O}_T(t)) = \bigoplus_{x \in T} \mathcal{O}_T(t)|_x,$$

the restriction of λ to the summand at x is zero if and only if $a(x) = 0$.

Proof. Write $i : T \hookrightarrow V$ and set $E = i_* \mathcal{O}_T(t)$. For $0 \leq j \leq r$, put

$$K_j = \mathcal{O}_V(t - jq)^{\oplus \binom{r}{j}}.$$

The ordered sections $\sigma_1, \dots, \sigma_r$ give the twisted Koszul resolution

$$(5.1) \quad 0 \longrightarrow K_r \longrightarrow \dots \longrightarrow K_0 \longrightarrow E \longrightarrow 0.$$

Indeed, at a point of T they generate the maximal ideal of a regular local ring of dimension r ; away from T one of them is a unit in a local trivialization. The leftmost term is

$$M = K_r = \mathcal{O}_V(t - rq) = \mathcal{O}_V(-r - 1).$$

All other terms have positive twists, since

$$t - jq = (r - j)q - r - 1 > 0 \quad (0 \leq j < r).$$

Their higher cohomology vanishes by assumption. The successive connecting maps therefore give an exact sequence

$$(5.2) \quad H^0(V, \mathcal{O}_V(t)) \longrightarrow H^0(T, \mathcal{O}_T(t)) \xrightarrow{\partial} H^r(V, M) \longrightarrow 0.$$

We show that, under duality, the transpose of ∂ is restriction of sections of $\omega_V(r + 1)$ to T . Let

$$\varepsilon \in \text{Ext}_V^r(E, M)$$

be the extension class of (5.1); its connecting homomorphism is ∂ . Proper duality gives

$$\begin{aligned} H^0(V, \omega_V(r + 1)) &\simeq \text{Hom}_V(M, \omega_V) \simeq H^r(V, M)^\vee, \\ \text{Ext}_V^r(E, \omega_V) &\simeq H^0(T, \mathcal{O}_T(t))^\vee. \end{aligned}$$

We regard a section a of $\omega_V(r + 1)$ as a morphism $a : M \rightarrow \omega_V$ via the first identification. The first line uses top-degree duality. For the second line, the canonical morphism $\omega_V[r] \rightarrow \omega_V^\bullet$ is an isomorphism near T . As E is supported on T , applying $R\mathcal{H}om_V(E, -)$ gives an isomorphism, and proper duality applies. Both duality identifications are compatible with morphisms and connecting maps [21, Tag 0FVV(5),(6)]. Hence $\partial^\vee$ is, up to the common sign convention, the map

$$\text{Hom}_V(M, \omega_V) \longrightarrow \text{Ext}_V^r(E, \omega_V), \quad a \longmapsto a_* \varepsilon,$$

obtained by pushing the extension along $a : M \rightarrow \omega_V$.

To compute this map, apply $\mathcal{H}om_V(-, \omega_V)$ to (5.1). Its final differential fits into the exact sequence

$$\omega_V(r + 1 - q)^{\oplus r} \xrightarrow{(\pm \sigma_1, \dots, \pm \sigma_r)} \omega_V(r + 1) \longrightarrow i_*(\omega_V(r + 1)|_T) \longrightarrow 0.$$

There are no other cohomology sheaves: near T this is the dual Koszul complex of a regular sequence on an invertible sheaf, and away from T the complex is contractible. It follows that

$$\mathrm{Ext}_V^r(E, \omega_V) \simeq H^0(T, \omega_V(r+1)|_T).$$

The class $a_*\varepsilon$ is represented by $a : K_r = M \rightarrow \omega_V$ in the last term of the dual complex, so its image is precisely $a|_T$. Dualizing (5.2) now proves the lifting assertion.

The identifications above commute with the direct-sum decomposition over the points of T . At each point they give a nondegenerate pairing between one-dimensional spaces. Consequently the component of the functional at x vanishes exactly when $a(x) = 0$. $\square$

The adjunction twist $rq - t = r + 1$ is independent of q . The following lemma embeds $\omega_V(r+1)$ into a fixed twist of $\mathcal{O}_V$.

Lemma 5.2. *Let $V \subseteq \mathbf{P}^n$ be integral of dimension r . There is an integer $m > 0$, depending only on this embedding, and an injection*

$$\omega_V(r+1) \hookrightarrow \mathcal{O}_V(m).$$

Proof. For $m \gg 0$, the sheaf $\mathcal{H}om_V(\omega_V(r+1), \mathcal{O}_V)(m)$ is globally generated. It has rank one, so a section nonzero at the generic point gives a generically injective morphism $\omega_V(r+1) \rightarrow \mathcal{O}_V(m)$. The kernel is torsion and hence zero, since $\omega_V(r+1)$ is torsion-free. $\square$

6. PROOF OF THE THEOREM

We first bound the number of points in a fiber on which a fixed-degree section can vanish.

Lemma 6.1. *Let $V \subseteq \mathbf{P}^n$ be integral of dimension $r \geq 1$ and degree δ . Let $g : V \rightarrow V$ be finite surjective, with $g^*\mathcal{O}_V(1) \simeq \mathcal{O}_V(q)$ for an integer $q \geq 1$. Suppose $y \in V_{\mathrm{reg}}$, g is étale over y , and $S = g^{-1}(y) \subseteq V_{\mathrm{reg}}$. If a nonzero section of $\mathcal{O}_V(m)$, $m > 0$, vanishes at every point of S , then $q \leq m\delta$.*

Proof. Let Z be the zero divisor of the section. It is an effective Cartier divisor, since V is integral. The projection formula gives $\deg g = q^r$, so S consists of q^r distinct points.

Pullbacks of target hyperplanes through y form a linear system in $|\mathcal{O}_V(q)|$ with finite base locus S . Choose $r-1$ general members $E_1, \dots, E_{r-1}$; for $r=1$ no members are needed. They meet Z properly at each successive intersection, because no positive-dimensional component is contained in the base locus. Thus $Z \cdot E_1 \cdots E_{r-1}$ is an effective zero-cycle. At each $x \in S$, the local equations of $Z, E_1, \dots, E_{r-1}$ form a system of parameters in the regular local ring $\mathcal{O}_{V,x}$. Their local intersection multiplicity is the positive length of the quotient by these equations. Each of the q^r points of S contributes at least one to the intersection cycle, so

$$q^r \leq \deg(Z \cdot E_1 \cdots E_{r-1}) = mq^{r-1}\delta;$$

see [13, Sections 2.3–2.4]. Dividing by q^{r-1} proves the claim. $\square$

Proof of Theorem 1.1. Write $q > 1$ for the algebraic degree of f , and put $r = \dim V$ and $\delta = \deg V$. If $r = 0$, then V is a point; if $r = n$, then $V = \mathbf{P}^n$. Assume $1 \leq r < n$ and suppose, for a contradiction, that $\delta > 1$.

Fix an injection $\omega_V(r+1) \hookrightarrow \mathcal{O}_V(m)$ as in Lemma 5.2. Replace f by an iterate, still denoted by f , whose algebraic degree q satisfies

$$q > \max\{r+1, m\delta\}.$$

Put $g = f|_V$ and $t = rq - r - 1$. By Lemma 2.1, there is a dense open set $U \subseteq V_{\text{reg}}$ such that $g^{-1}(U) \subseteq V_{\text{reg}}$ and $g^{-1}(U) \rightarrow U$ is étale.

A general codimension- r linear subspace meets V transversely in δ distinct points $y_1, \dots, y_\delta$ of U . Indeed, it avoids $V \setminus U$ by dimension, and Bertini's theorem gives transversality on the smooth locus. Set

$$S_j = g^{-1}(y_j), \quad T = S_1 \sqcup \dots \sqcup S_\delta.$$

Each S_j consists of q^r distinct smooth points. Pulling back the r linear equations of the chosen subspace gives sections of $\mathcal{O}_V(q)$ with common zero scheme T . This zero scheme is reduced because g is étale over the transverse intersection. By Corollary 2.3, the cohomology hypothesis of Proposition 5.1 holds.

Proposition 4.1 gives a functional on $H^0(S_1, \mathcal{O}_{S_1}(t))$ with nonzero component at every point of S_1 , annihilating all restricted global sections of $\mathcal{O}_V(t)$. Extend it by zero on $T \setminus S_1$. Proposition 5.1 then gives a section $a \in H^0(V, \omega_V(r+1))$ satisfying

$$a(x) \neq 0 \quad (x \in S_1), \quad a(x) = 0 \quad (x \in T \setminus S_1).$$

In particular, a is nonzero and vanishes on S_2 , which exists because $\delta > 1$. Its image in $H^0(V, \mathcal{O}_V(m))$ is still nonzero and still vanishes on S_2 . Lemma 6.1 gives $q \leq m\delta$, contrary to the choice of q .

Therefore $\deg V = 1$. If $\mathbf{P}^s$ is the linear span of V , the degree bound for a nondegenerate integral variety gives

$$1 = \deg V \geq s - r + 1.$$

Thus $s = r$, and the closed integral subvariety $V \subseteq \mathbf{P}^r$ of dimension r is all of $\mathbf{P}^r$. $\square$

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